

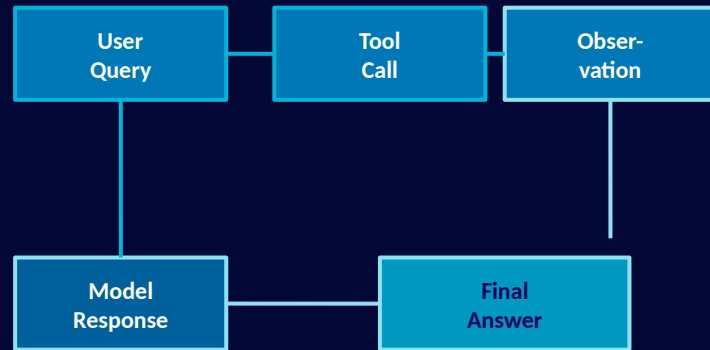
DiaTool-DPO

Multi-Turn Direct Preference Optimization

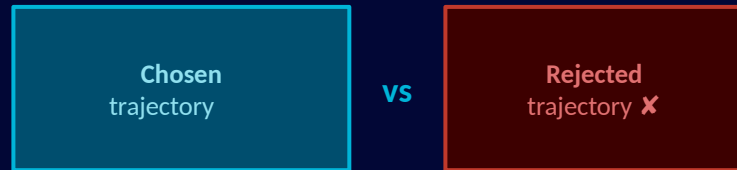
for Tool-Augmented LLMs

Automatic preference pair construction • 5-state MDP • Zero human labeling

arxiv:2504.02882



DPO Preference Pairs



Automatic • No human labeling • 5-state MDP



Tool-Augmented LLM Dialogue Flow

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

01 Hallucinated Tool Invocations

SFT models call tools even when arguments are missing, fabricating slot values

02 Missing Follow-Up Questions

Model fails to ask for missing required parameters before calling a tool

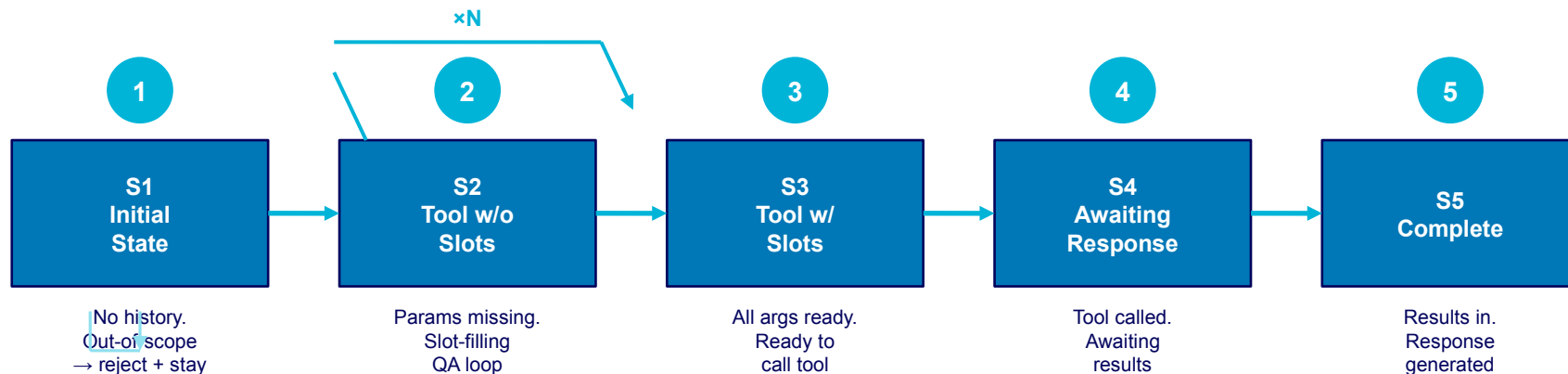
03 Failed Tool Rejection

Model attempts to call non-existent tools for out-of-scope requests

Why SFT Falls Short

- SFT learns from expert trajectories only
- No signal on what NOT to do
- Cannot model preference between correct and incorrect dialogue flows

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



Dialogue Trajectory Types

Type 1 (complete query)

$S1 \rightarrow S3 \rightarrow S4 \rightarrow S5$

Type 2 (incomplete query)

$S1 \rightarrow (S2 \times N) \rightarrow S3 \rightarrow S4 \rightarrow S5$

Type 3 (out-of-scope)

$S1 \rightarrow S1$ (rejection + remain in initial state)

MDP Formulation

States: $S = \{S1, S2, S3, S4, S5\}$

Actions: reject / slot-fill / call tool / respond

Transitions: deterministic given system state

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Scenario	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	All required arguments present in query	Book a flight from NYC to LA on May 5th
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Missing arguments; slot-filling loop repeats N times	"Book a flight" [missing date, destination]
Type 3	$1 \rightarrow 1$	Requested functionality unavailable; tool call rejected	"Translate this text to Latin" [no translation tool]

Type 1

No slot-filling needed

Type 2

N slot-filling rounds until complete

Type 3

Rejection prevents hallucinated tool calls

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089 (Easy)
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562 (Hard)
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530 (Hard)
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089 (Easy)
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562 (Hard)
Type 3	1→1	1→3→4	Tool call reject	567 (Hard)
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562 (Hard)

Zero Human Labeling

Chosen/rejected pairs constructed automatically from MDP state definitions

Covers All Failure Modes

Redundant slot-filling, slot hallucination, and tool rejection each get dedicated pairs

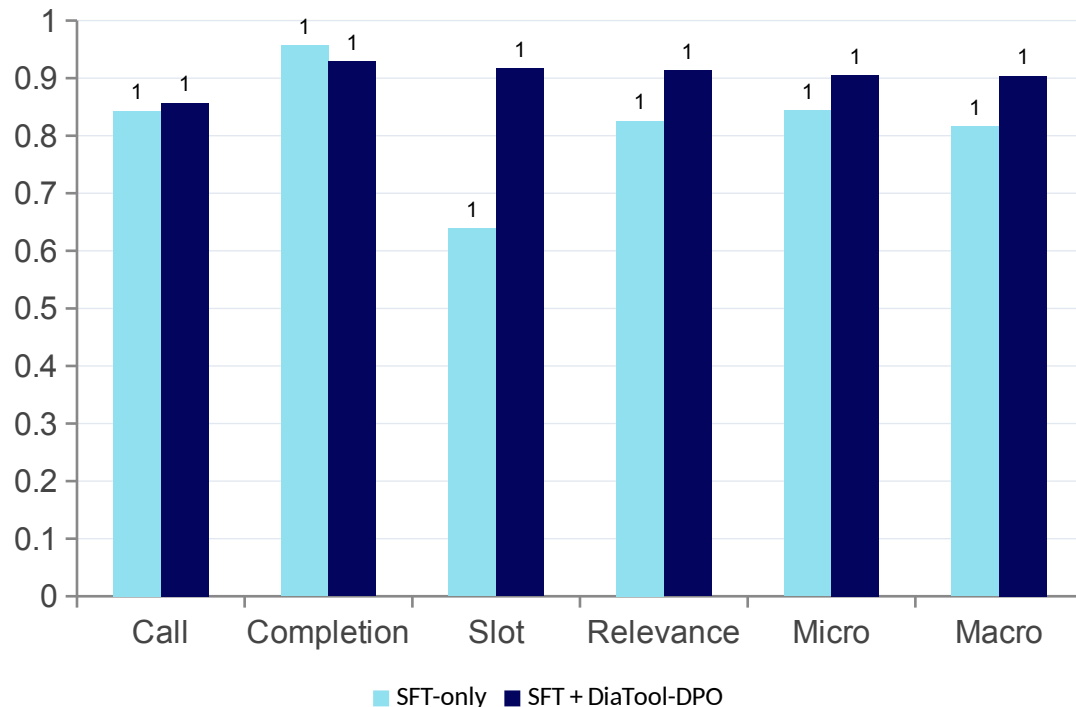
Easy + Hard Variants

Both easy (single deviation) and hard (multi-step) negative examples included

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Compl.
FunctionChat-Bench	Baseline	0.314	0.700
	SFT + DiaTool-DPO	0.900	0.916
	SFT + DiaTool-DPO		
★ SFT + DiaTool-DPO achieves highest Macro Avg for all model sizes Eval: FunctionChat-Bench			

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



+43.5%

Slot Accuracy Gain

+10.5%

Relevance Gain

0.639 → 0.917

Slot: SFT-only → SFT+DPO

0.826 → 0.913

Relevance: SFT-only → SFT+DPO

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1 MDP formulation enables systematic trajectory pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected pairs without human annotation

2 DPO complements SFT — not a replacement

DiaTool-DPO alone performs poorly; it must be combined with SFT for best results

3 Slot-filling and tool rejection see the largest gains

LLaMA-3-8B: Slot accuracy +43.5%, Relevance +10.5% over SFT-only

4 SFT + DiaTool-DPO achieves best Macro Average across all model sizes

Consistent improvement for 3.1B and 8B parameter models

5 Language-agnostic and production-ready

Method generalizes beyond Korean to English; opens path to near GPT-4o performance (94.8% info gathering, 91% rejection) on smaller models